Bulk Milk Cooler



Category: **Animal Husbandry & Dairy** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Bulk milk Cooler

What it is used for

A bulk milk cooler rapidly cools and stores freshly collected milk at a controlled temperature to slow bacterial growth and preserve quality before collection or processing. Operate it using hygienic milk-handling practices, monitor cooling performance, and follow applicable dairy requirements for temperature and storage time.

Main parts

- Insulated milk tank and lid
- Cooling compressor and refrigeration circuit
- Agitator or milk-stirring system
- Temperature sensor and display
- Milk inlet, outlet valve, and drain
- Cleaning system, control panel, and electrical protection

Before you start

1. Confirm the tank capacity, cooling specification, and required milk storage temperature from the dairy procedure.
2. Inspect the tank, lid, seals, valves, agitator, refrigeration unit, and electrical connections for damage or contamination.
3. Verify that the tank and milk-contact surfaces have been cleaned and sanitised according to the approved procedure.
4. Check that the cooling system, agitator, temperature display, and alarms operate correctly.
5. Ensure the milk inlet and outlet connections are clean, capped when not in use, and protected from contamination.
6. Confirm that the tank is positioned on a stable surface with adequate ventilation around the refrigeration unit.

How to use

1. Switch on the cooling system before or at milk intake according to the approved operating procedure.
2. Verify the displayed temperature and confirm that the tank is ready to receive milk.
3. Transfer freshly collected milk through the clean inlet using hygienic handling procedures.
4. Close the lid promptly and start the agitator according to the tank manufacturer's instructions.
5. Monitor milk temperature, cooling time, agitator operation, and any alarms during the cooling cycle.

6. Confirm that the milk reaches and remains within the temperature required by the dairy procedure and applicable regulations.
7. Avoid unnecessary lid opening and prevent water, cleaning chemicals, or foreign material from entering the tank.
8. Record milk intake time, quantity, temperature readings, and any abnormal cooling delays.
9. Before milk collection, follow the approved procedure for agitation, sampling, and transfer to the collection vehicle or processing line.
10. After emptying the tank, isolate power as required and carry out the prescribed wash, rinse, and sanitising cycle.
11. Inspect the tank and valves after cleaning, allow surfaces to drain as specified, and record cleaning completion or faults.

⚠ Safety

- Only trained personnel should operate, clean, or maintain the cooler.
- Never enter the tank or reach into it while the agitator is energised; isolate and verify power before internal cleaning or maintenance.
- Keep electrical connections dry and use the installed earthing and protective devices.
- Use approved detergents and sanitising agents at the specified concentration; never mix incompatible chemicals.
- Treat refrigerant systems and pressure components as specialist service items; do not open or repair them without qualification.
- If cooling fails or milk temperature exceeds the permitted limit, notify the responsible dairy operator and follow the milk quality disposition procedure.

Care & storage

- Clean and sanitise all milk-contact surfaces after each batch or as required by the dairy schedule.
- Inspect seals, valves, agitator blades, sensors, and the tank outlet regularly.
- Check refrigeration performance, condenser cleanliness, and airflow around the unit.
- Calibrate or verify the temperature sensor at the specified intervals.
- Arrange qualified servicing for refrigerant leaks, abnormal noise, poor cooling, or repeated alarms.

Common mistakes to avoid

- Loading milk into an inadequately cleaned or sanitised tank.
- Ignoring slow cooling or incorrect temperature readings.
- Leaving the lid open or allowing contaminated equipment to contact milk.
- Cleaning or servicing the agitator without isolating power.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.